

Material Properties

- Photoelectric Effect: the emission of an electron when electromagnetic radiation hits some material.
 - Work Function (Φ): minimum energy required to eject electron from material in [Joules]

$$E_k = hf - \Phi$$

maximum possible
ejection energy [J] $\uparrow$ energy of photon [J]

- relationship between Momentum (p) & Wavelength (λ):

$$p = \frac{h}{\lambda} \quad \leftarrow \text{planck's constant}$$

- Wave Nature of Matter

- all matter exhibits both particle and wave like behaviour
- can be observed with an electron in a double-slit experiment

- de Broglie Wavelength: the wavelength of a particle with mass: $\lambda = \frac{h}{p}$

- Heisenberg Uncertainty Principle: the more precisely the **position** of some particle is known, the less precisely its **momentum** can be predicted from initial conditions, and vice versa

$$\sigma_x \cdot \sigma_p \geq \frac{h}{4\pi}$$

standard deviation of position $\uparrow$ standard deviation of momentum

- Schrodinger's Equation: describes wave function of quantum-mechanical system.

$$j \cdot \hbar \cdot \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \cdot \frac{\partial^2 \Psi}{\partial x^2} + U \cdot \Psi$$

reduced Planck's constant: $\hbar = \frac{h}{2\pi}$ $\uparrow$ potential of system $\uparrow$ Wave Function

- the probability of finding a particle somewhere in space can be determined by:

$$P_R = \int_R |\Psi|^2 d\tau; \quad \text{where } R \text{ is a region of space}$$

- Normalized Condition: 100% chance of particle being somewhere: $P_{\infty} = \int |\Psi|^2 d\tau = 1$

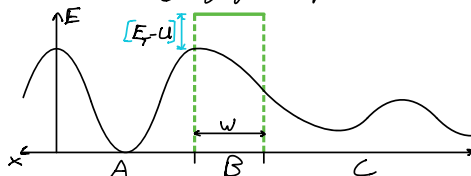
- Time Independent Equation

$$\frac{\partial^2 \Psi(x)}{\partial x^2} = -\frac{2m}{\hbar^2} [E_T - U(x)] \Psi(x)$$

$\uparrow$ total energy of particle

- Quantum Tunnelling: a particle's wave function can propagate through a **potential barrier** $\rightarrow E_T < U$

- effect is only significant for $w < 10\text{nm}$

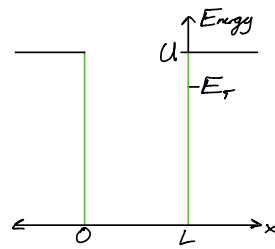


Wave Equations:

$$A/C: \frac{\partial^2 \Psi(x)}{\partial x^2} = -\frac{2m E_T}{\hbar^2} \cdot \Psi(x)$$

$$B: \frac{\partial^2 \Psi(x)}{\partial x^2} = -\frac{2m [E_T - U(x)]}{\hbar^2} \cdot \Psi(x)$$

- Finite Potential Well: a particle is trapped within *potential walls* but can escape
 - similar to Quantum Tunneling, the particle's wave function can penetrate these *potential walls*



$$\Psi(x) = A e^{\sqrt{\frac{2m}{\hbar^2}(U-E_T)}x}, \quad x < 0$$

$$\Psi(x) = B e^{-\sqrt{\frac{2m}{\hbar^2}(U-E_T)}x}, \quad x > L$$

- as U approaches infinity, Ψ approaches 0

- Infinite Potential Well: a particle is trapped within potential walls infinitely high
 - the wave function will be 0 anywhere in the well

$$\Psi(x, t) = A \sin\left(\sqrt{\frac{2mE}{\hbar^2}}x\right) e^{j\frac{E}{\hbar}t} \rightarrow \text{represents 2 waves propagating in opposite directions within the well}$$

- all energy within the well is kinetic: $E_k(n) = \frac{\hbar^2 k^2}{2m}$, $k = \frac{n\pi}{L}$; $n \in \mathbb{Z}$

- Amplitude of Wave Equation:

$$\Psi(x) = \sqrt{\frac{2}{L}} \sin(kx)$$

Spatial Frequency

- use *normalized condition* $\rightarrow$ 100% chance that particle is in the well

- Multi-Dimensional Confinement: confining a particle to a space with dimensions of its de Broglie Wavelength
 - allows for the observation of a particle's quantum mechanical motion

$$\text{1D: } E_k(n_x) = \frac{\hbar^2}{8m^*} \cdot \frac{n_x^2}{L_x^2}$$

- average separation of states *increases* with energy

$$\text{2D: } E_k(n_x, n_y) = \frac{\hbar^2}{8m^*} \left(\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} \right)$$

- average separation of states *does not change* with energy

$$\text{3D: } E_k(n_x, n_y, n_z) = \frac{\hbar^2}{8m^*} \left(\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2} \right)$$

- average separation of states *decreases* with energy

- Degenerate Energy Levels

level is degenerate if it corresponds to more than one measurable state

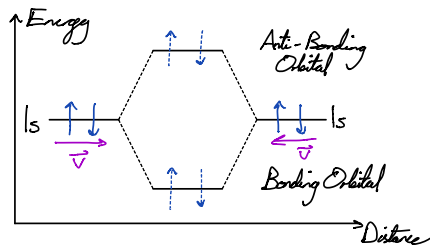
- Degeneracy: number of states at certain energy level

Atoms & Solids

- electron **orbitals** are standing waves of spherical symmetry

Bonding

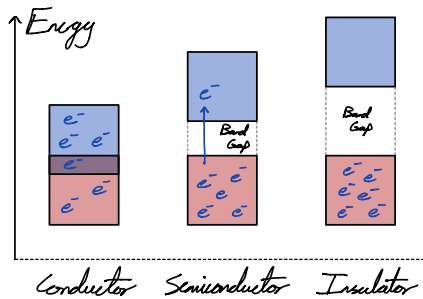
- When atoms come close together, each orbital splits into a **bonding** and an **anti-bonding** state (N orbitals will create N new states)



- if the **electrons** from the two $1s$ orbitals fill the bonding and anti-bonding states, **bonding fails**

Bands

- when many atoms come together, the electrons occupy a band of energy levels
- Conduction Band**: electrons **enter into** from the Valence Band
- Valence Band**: electrons **leave to** the Conduction Band
- Bandgap**: energy **gap** between Valence and Conduction Bands

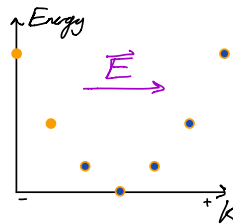
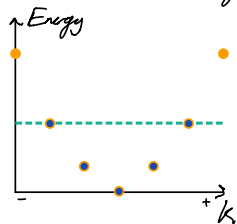


- larger bandgap decreases electron crossover between **Conduction** and **Valence** Bands
- there must be space in the Bands for electrons to move; for current to flow

- Band properties of materials determine their electronic characteristics

- Spin**: rotation of electrons that is predicted by relativistic quantum mechanics
 - each orbital contains a **spin-up** and **spin-down** electron

- Fermi level**: the highest filled electron level

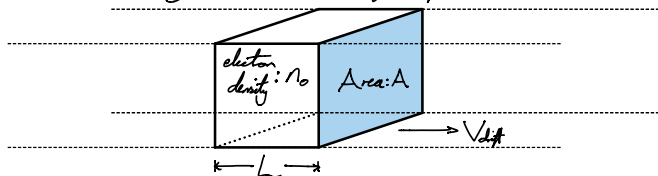


- Electric Field** generates **electron** flow (current)

- Conduction**: the movement of **electrons** through a medium
 - mobility** is limited by electron collisions and thermal vibrations ($v = \mu E$)

— Semiconductors

- material with modifiable conductivity value between that of conductors and insulators
- Crystalline Solids have a regular and well-defined atomic structure
 - they experience different properties in different directions
- Polycrystalline Material: multiple crystalline structures separated by some domains
- Charge Carriers
 - free to move particle that carries a charge
 - Electrons $\rightarrow$ negatively charged moving electrons
 - Holes $\rightarrow$ positively charged moving representations of missing electrons
- Intrinsic Material
 - pure material with minimal electron excitation to the conduction band
 - intrinsic electron concentration equals intrinsic hole concentration: $n_i = p_i$
 - temperature and size of bandgap can alter excitation probability
- Extrinsic Material
 - intrinsic material doped with different elements to alter electric properties of material
 - Donors: dopant atom with extra electron (e.g. Phosphorus); can form n-type region
 - Acceptors: dopant atom missing electron (e.g. Aluminium); can form p-type region
- Effective Mass (m^*)
 - a particle responding to force from nearby identical particles; this force can be interpreted as a change in mass
 - depends on lattice material of crystalline solid
- Current
 - moving stream of charged particles: $\frac{dQ}{dt}$



- charge density takes time (τ) to move distance L :

$$L = v_{\text{drift}} \cdot \tau$$

- charge within box: $Q = n_0 \cdot A \cdot L \cdot q_e$ charge of an electron

$$I = n_0 \cdot A \cdot q_e \cdot v_{\text{drift}} \rightarrow J = n_0 \cdot q_e \cdot v_{\text{drift}}$$

- Carrier Mobility (μ_e / μ_h)
 - property describing how quickly a carrier can move due to electric field

$$\mu = \frac{q_e}{m^*} \cdot \tau_{\text{collision}} \quad \leftarrow \text{average time between carrier collisions}$$

- Conductivity: $\sigma = n_0 \cdot q_e \cdot \mu_e$
 - electron speed changes depending on material conductivity
- Ohm's Law: $R = \frac{L}{\sigma A}$; the degree to which current is resisted depends on material conductivity
- Phonons: quantum mechanical description of lattice vibrations
 - electron impedance increases with more phonons
- rise in temperature increases number of carriers $\rightarrow$ increase conductivity
- mobility decreases with rise in temperature due to the rich lattice vibrations intercepting electrons
- Diffusion: carrier movement due to electric field generated by non uniform carrier concentration
- Dynamic Equilibrium — Law of Mass Action
 - Generation: electron gains energy and moves from the valence to conduction band
 - Recombination: electron loses energy and moves from the conduction to valence band
 - at Thermal Equilibrium the rate of generation equals the rate of recombination resulting in a constant electron and hole concentrations.

$$p_i^2 = n_i^2 = n_0 p_0$$

- Charge Balance
an electrically neutral crystal at room temperature possesses the following property

$$N_D^+ + p_0 = N_A^- + n_0$$

- calculating total Electron concentration (n_0) and total Hole concentration (p_0)

N_D : electron donor density

N_D^+ : donors with missing electrons density

N_A : electron acceptor density

N_A^- : acceptors with electrons density

- N-Type material: $N_D^+ > N_A^- \rightarrow$ electrons are majority carrier

- P-Type material: $N_A^- > N_D^+ \rightarrow$ holes are majority carrier

- using equations $n_i^2 = n_0 p_0$ and $N_D^+ + p_0 = N_A^- + n_0$ to get the quadratic equations:

$$\left. \begin{aligned} n_0^2 + (N_A^- - N_D^+) n_0 - n_i^2 &= 0 \\ p_0^2 + (N_D^+ - N_A^-) p_0 - n_i^2 &= 0 \end{aligned} \right\} \text{the solutions of these equations give } n_0 \text{ \& } p_0$$

- if $N_A^- \gg n_i \rightarrow p_0 \approx N_A^- - N_D^+$ or if $N_D^+ \gg n_i \rightarrow n_0 \approx N_D^+ - N_A^-$

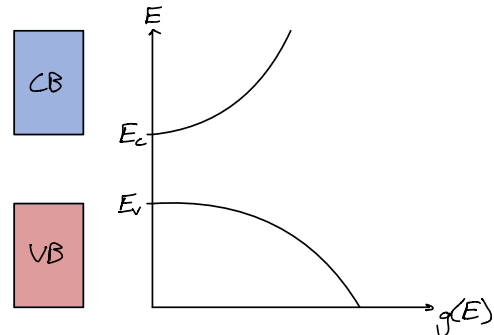
- Density of States: $g(E)$
 - number of states per volume at energy E within the interval dE

$$g_c(E) = \frac{8\pi\sqrt{2}}{h^3} m_e^{*3/2} \sqrt{E - E_c}$$

DoS in conduction band

$$g_v(E) = \frac{8\pi\sqrt{2}}{h^3} m_e^{*3/2} \sqrt{E_v - E}$$

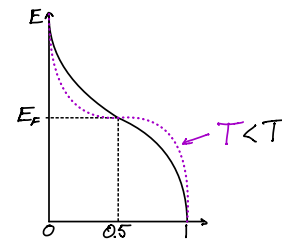
DoS in valence band



- Occupancy
 - Fermi Level (E_F): energy level with 50% chance of occupancy
 - when close to VB $\rightarrow$ lots of holes: *p-type*
 - when close to CB $\rightarrow$ lots of electrons: *n-type*
 - middle of VB and CB $\rightarrow$ *intrinsic material*

- Fermi-Dirac Distribution: $f(E) = \left[1 + e^{\frac{E - E_F}{kT}}\right]^{-1}$

- Pauli Exclusion Principle
 - no 2 electrons can occupy the same energy state or spin state
 - Fermions obey this principle



- Current Density: $J = qE(n_0\mu_e + p_0\mu_h)$

- Counting Electrons

$$n_0 = \int_{E_c}^{\infty} g_c(E) f(E) dE$$

$$p_0 = \int_{-\infty}^{E_v} g_v(E) [1 - f(E)] dE$$

- discrete band of states can be treated as continuous due to high number of states

- Effective Density of States (N_c^* , N_v^*)
 - the density of states near edge of the conduction (E_c) or valence (E_v) band

$$n_0 = N_c^* \cdot e^{-\frac{E_c - E_F}{kT}}$$

$$N_c^* = 2 \left[\frac{2\pi m_e^* kT}{h^2} \right]^{3/2}$$

$$p_0 = N_v^* \cdot e^{-\frac{E_F - E_v}{kT}}$$

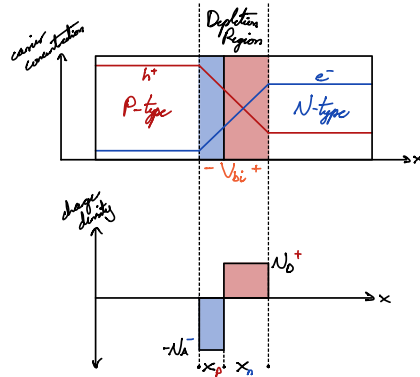
$$N_v^* = 2 \left[\frac{2\pi m_h^* kT}{h^2} \right]^{3/2}$$

- Determining Fermi Level

- in Intrinsic Material: $E_F = \frac{E_v + E_c}{2} + \frac{kT}{2} \ln \left(\frac{N_v^*}{N_c^*} \right)$

PN Junctions

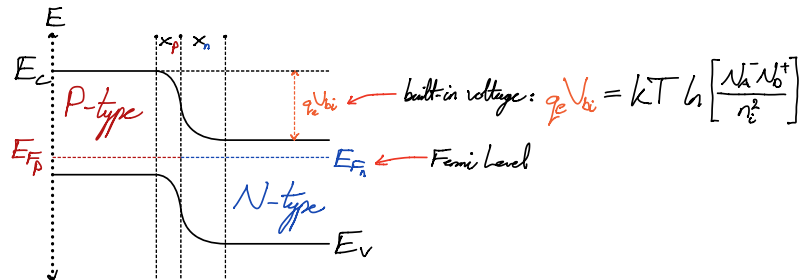
- **P-type** and **N-type** material brought close together
- Depletion Region: at boundary between P and N type material, some electrons and holes diffuse across, and then undergo recombination



- electron recombination in **P-type** material creates **negatively** charged ions
- hole recombination in **N-type** material creates **positively** charged ions
- Conservation of Charge:

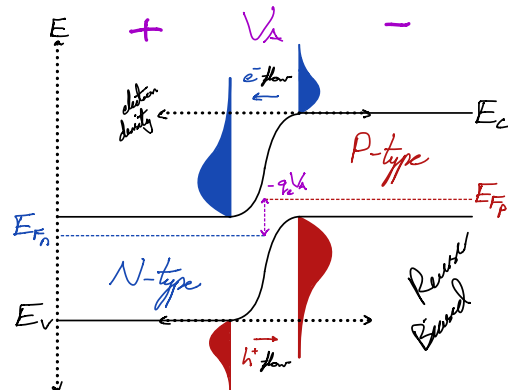
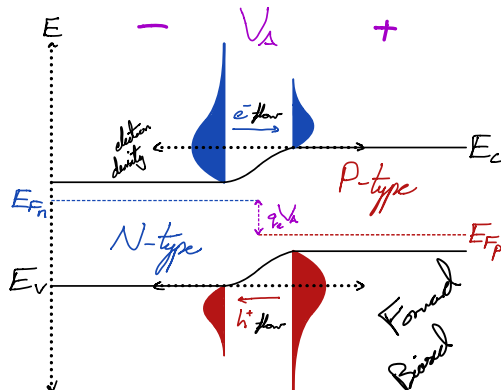
$$x_n \cdot N_D^+ = x_p \cdot N_A^-$$

- these models use the **Depletion Approximation**
- Band Bending: the bending of energy states through the depletion region



Biasing of a PN Junction

- apply **voltage** across junction to allow current flow
- Quasi Fermi Level: shifted Fermi level due to hole and electron concentration out of equilibrium



• Ideal Diode Equation

- carrier concentration near depletion region depends on applied voltage

$$n(x_p) = n_{op} \cdot e^{\frac{qV}{kT}} \quad p(x_n) = p_{on} \cdot e^{\frac{qV}{kT}}$$

- concentration decays towards edges of depletion region

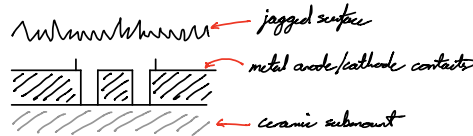
$$I = I_0 \left[e^{\frac{qV}{kT}} - 1 \right]$$

↑
exchange current

- electrons can tunnel from valence to conduction band in strong reverse bias

• Diode Applications

- LEDs: recombination during forward bias releases energy as photons
- roughened surface decreases internally reflected light



- Solar Cells: photons knock electrons up into conduction band generating current
 - ideal cell material possesses bandgap similar to energy of photon
 - excess energy from photon is wasted as heat
 - multiple materials with different bandgaps increases range of emission capture

Dielectrics

- electrical insulator that can be polarized by an applied field
- Insulator: material with large bandgap

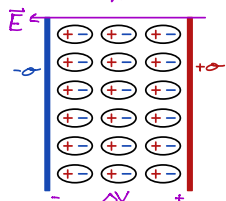
- Crystalline Dielectrics: $E_g > 2eV$

- Non-Crystalline Dielectrics

- Ex: Plastic: very low number of free electrons

- Polarization ($\vec{P}$): sum of dipole moments per volume

- controls applied field by decreasing voltage across dielectric $\rightarrow$ lower conductivity



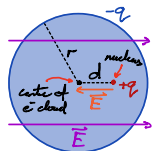
charge due to polarization

charge on electrodes

$$\sigma_{pe} = \sigma_{bound} + \sigma_{vacuum} = \vec{P} \cdot \hat{n} + \epsilon_0 |\vec{E}| = (\chi + 1) \epsilon_0 |\vec{E}|$$

↑
susceptibility

- coulombic force cancels out applied field force at equilibrium



$$\vec{E} = -\vec{E} = \frac{qd}{4\pi\epsilon_0 r^3}$$

distance at equilibrium

- Dipole Moment: $\vec{p} = qd = 4\pi\epsilon_0 r^3 \vec{E} = \alpha \vec{E}$ atomic polarizability
- Types of Polarization
 - different types of polarization occur at certain frequency ranges

1. Electronic Polarization

- dipole of electron cloud and nucleus generated by $\vec{E}$

2. Ionic Polarization

- lattice ions are stretched by $\vec{E}$ increasing lattice dipole moment
- operates like mass-spring system:

$$mx'' = q\vec{E} - kx - bx' \xrightarrow{L} \frac{p(s)}{E(s)} = \frac{q^2}{ms^2 + bs + k}$$

3. Orientational Polarization

- $\vec{E}$ generates torque on molecule dipole
- depends on ratio of dipole energy to thermal energy:

$$\vec{p} = \frac{\rho^2 \vec{E}}{kT}$$

- Debye Equation: dielectric constant at orientational frequencies

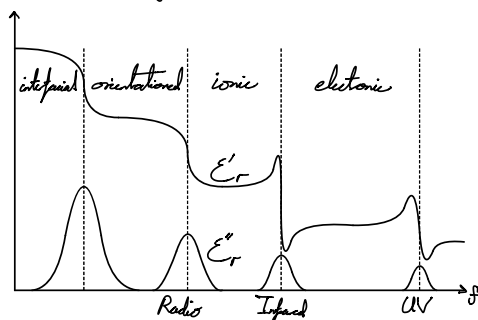
$$\epsilon' + j\epsilon'' = \frac{\epsilon_s - \epsilon_\infty}{1 + j\omega\tau} + \epsilon_\infty$$

static ϵ_r ← ϵ_r when applied period is greater than alignment time

4. Interfacial Polarization

- charge builds at opposing ends of dielectric due to $\vec{E}$

- Relative Permittivity: $\epsilon_r = \epsilon'_r - j\epsilon''_r$



ϵ'_r : represents degree of polarization

ϵ''_r : associated with dielectric losses

- spikes of imaginary permittivity represent high loss $\rightarrow$ resonant frequencies

- Static ϵ_r (ϵ_s): relative permittivity at a frequency of 0Hz

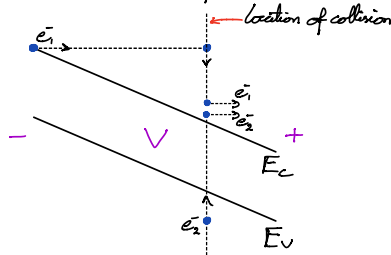
- Breakdown
 - sudden increase in current at high voltage

1. Zener Breakdown

- electrons tunnel from *p-type* VB to *n-type* CB during *extreme reverse bias*

2. Avalanche Breakdown

- compounding effect of CB electrons promoting VB electrons through *collisions* during reverse bias



3. Thermal Breakdown

- compounding effect of heat from *resistive losses* promoting electrons $\rightarrow$ increasing current

4. Discharge Breakdown

- *gas* within material ionises and damages material

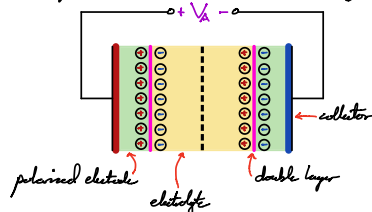
• Applications

• Electrolytics

- dielectric is composed of an ionic medium with a thin oxide layer
- possesses very high capacitance; provides low-frequency control

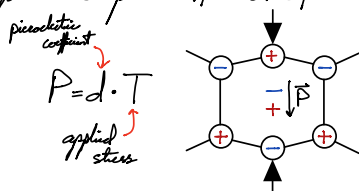
• Electrostatic Double-layer Capacitor (Supercapacitor)

- liquid dielectric forms double layer of charge at surface of electrode
- possesses high capacitance and large energy storage



Piezoelectrics

- electric charge generated as a result of mechanical stress
- upon compression/tension of material, dipole moment is created:



- polarisation occurs in multiple directions:

$$\begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = \begin{bmatrix} d_{xx} & d_{xy} & d_{xz} \\ d_{yx} & d_{yy} & d_{yz} \\ d_{zx} & d_{yz} & d_{zz} \end{bmatrix} \begin{bmatrix} T_x \\ T_y \\ T_z \end{bmatrix}; \text{ stress in normal direction}$$

• Direct Piezoelectric Effect

- surface charge generated by polarization: $\sigma_{\text{bound}} = \vec{P} \cdot \hat{n}$

• Inverse Piezoelectric Effect

- applied voltage actuates material through strain:

$$\begin{bmatrix} S_x \\ S_y \\ S_z \end{bmatrix} = \begin{bmatrix} d_{xx} & d_{xy} & d_{xz} \\ d_{xy} & d_{yy} & d_{yz} \\ d_{xz} & d_{yz} & d_{zz} \end{bmatrix} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix} \quad \text{applied field}$$

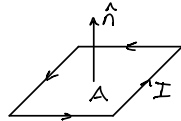
$$S = d \cdot E = \frac{\Delta L}{L}$$

- K-factor: K^2 represents mechanical energy converted to electrical energy

Magnetic Materials

- Ferromagnets: magnetic field generated by electron spins retaining previous alignment
- Paramagnets: electron spins align with external field to *boost* field
- Diamagnets: electron spins align with external field to *diminish* field

- Magnetic Dipole Moment: $\vec{m} = A \cdot \vec{I} \cdot \hat{n}$



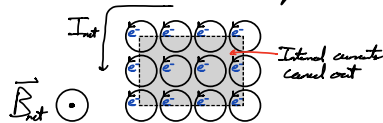
$$\vec{\tau} = \vec{m} \times \vec{B} \rightarrow \tau = AIB \sin \theta$$

↑ external field

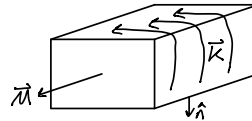
- Magnetization: average magnetic moment per volume: $\vec{M} = \frac{\sum_k \vec{m}_k}{V}$

• Equivalent Surface Current

- net current of electron spins is non-zero at surface of magnetized solid



- multiple layers of this electron slice represents a solid of magnetized material:

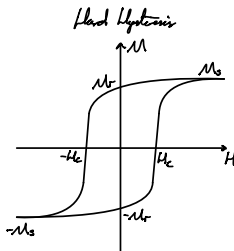


- current at surface per unit length ($\vec{K}$):

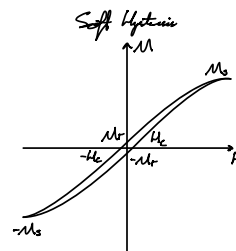
$$\vec{K} = \vec{M} \times \hat{n}$$

- combined current of magnetization and conduction: $I_{\text{total}} = \mu_0 M + \oint \vec{H} \cdot d\vec{l}$

• Hard versus Soft magnetic materials



- large Remanence
- large Coercivity



- small Remanence
- small Coercivity

- Remanence: B/H field that remains when field (H) is removed
- Coercivity: external field necessary to eliminate remanence (H_c)

- Anisotropy: dependency of magnetic properties on orientation of material
 - certain orientations cause solid to resist magnetization due to internal field
- Barkhausen Effect
 - domains of polarized material will "snap" to alignment under external field
 - hysteresis curve presents micro-steps as applied field increases